L5

CLASS X - SCIENCE

ACIDS, BASES AND SALTS

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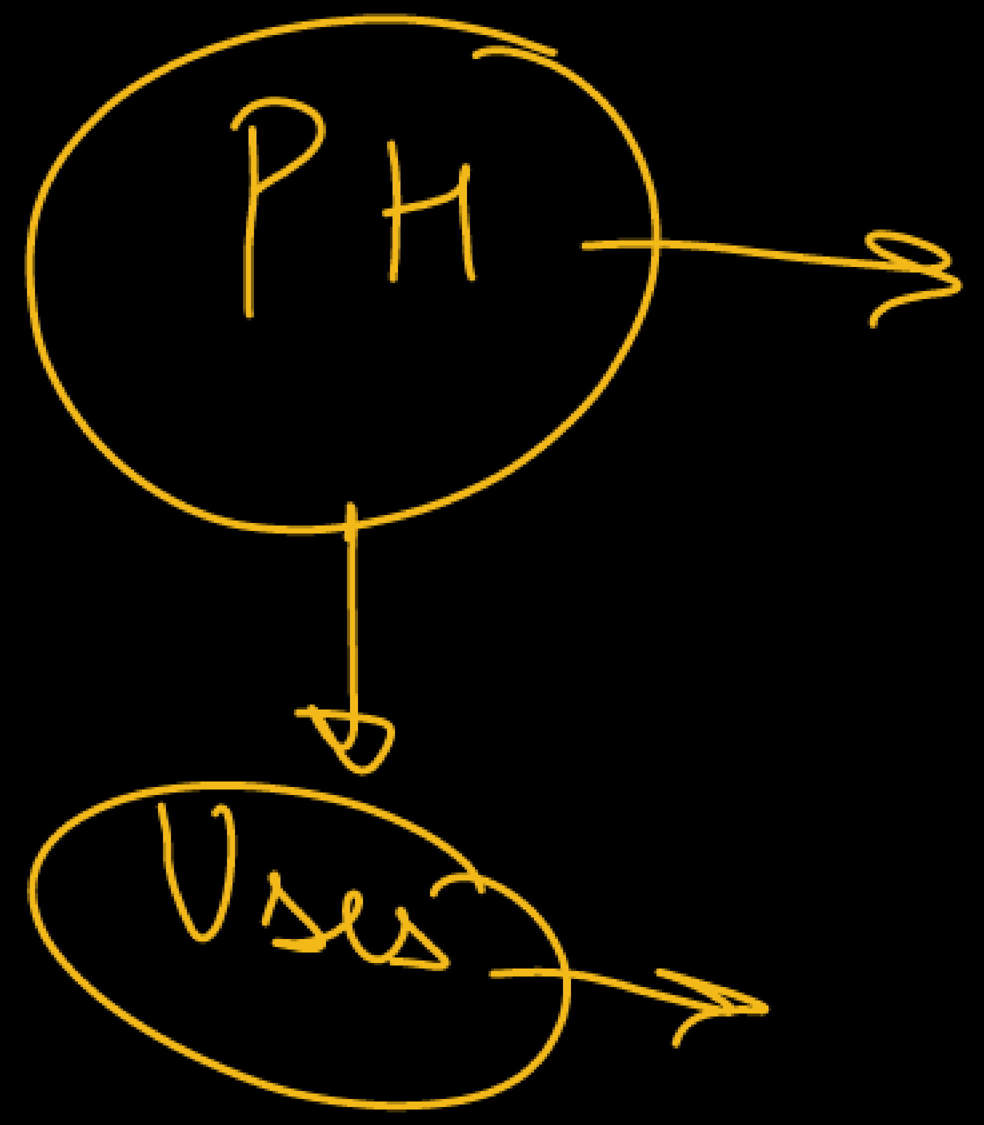

PH
Uses

Soch ke bataoo

Q. What is meant by the term pH of a solution? The pH of rain water collected from two cities 'A' and 'B' was found to be 6.1 and 5.3 respectively. The water of which city is more acidic? Justify your answer. State with reason what would happen to the aquatic life of a pond in which the rain water of city 'B' flows.

Soch ke bataoo

Q. A milkman adds a very small amount of baking soda to fresh milk.

(A) Why does he shift the pH of the fresh milk from 6 to slightly alkaline?

(B) Why does this milk take a long time to set as curd?

7-8

Soch ke bataoo

Q. Ritvik and his classmates were given an assignment for the summer vacations by his science teacher. They were told to collect samples of any 5 common substances for testing their pH and recording their nature accordingly. Ritvik scouted his house for different substances to complete the assignment and recorded his observations in the given table:

S. No.	Solution of the common substance	Colour of pH paper	pH value (Approx.)	Nature of substance
(1)	Lemon juice	Red	2.2 – 2.4	Acidic
(2)	Carrot juice	Green	6.1 – 7.0	Slightly acidic
(3)	Coffee	Orange	4.5 – 5.5	Slightly acidic
(4)	Colourless aerated drink	Red	3.0 – 3.5	Acidic
(5)	Saliva (after meal)	Orange	5.1 – 5.5	Slightly acidic

(A) Which of the given substance will retain its characteristic smell if Ritvik adds it to the lemon juice?

(a) Vanilla extract

(b) Sunflower oil

(c) Groundnut oil

Give justification for your answer.

Substance	pH
Battery acid	< 1.0
Stomach acid	2
Lemon juice	2.4
Cola	2.5
Apple juice	3.5
Black coffee	5
Black tea	5.5
Acid rain	5.6
Milk	6.5
Distilled water	7
Human saliva	7.5
Sea water	8
Soap	9.0 – 10.0
Milk of magnesia	10.5
Ammonia	11.5
Bleach	12.5

Acid → pH ↑

B) Ritvik came across an interesting fact during this assignment, where he learned that our stomach produces HCl. He wanted to know whether indigestion is related to this or not. Give your answer with an appropriate reason. Also provide a commonly used chemical substance to treat this problem.

→ $Mg(OH)_2$ ✓

OR

(B) When Ritvik measured the pH of saliva (before meal), he found it to be around 7.4 (neutral). However, the pH of saliva (after meal) was in the range of 5.1 – 5.5 (slightly acidic). Why is there such a shift in the pH value before and after meal? Does this change depend upon the food we eat? Explain briefly.

(C) What do you understand by the term synthetic indicators? Which of the following is a synthetic indicator combined with its correct colour changing characteristic in a basic medium? Provide an appropriate reason for your answer.

(a) Methyl orange; Yellow
(b) Turmeric extract; Green
(c) Clove oil; Blue
(d) Cabbage extract; Colourless

Soch ke bataoo

Q. The pH is quite useful to us in a number of ways in daily life. Some of its applications are:Control of pH of the soil : Plants need a specific pH range for proper growth. The soil may be acidic, basic or neutral depending upon the relative concentration of H* and OH-. The pH of any soil can be determined by using pH paper. If the soil is too acidic, it can be corrected by adding lime to it. If the soil is too basic, it can be corrected by adding organic manure which contains acidic materials.

H^{+}

(i) If the concentration of H_3O^+ ions increases in a solution, what will happen to its pH value?

(a) pH will increase

(b) pH will decrease

(c) pH will become 7

(d) pH will remain same

Acid ↑

pH ↓

11 12 13

2 → 3 → 4

(ii) P is an aqueous solution of acid and Q is an aqueous solution of base. When these two are diluted separately, then

(a) pH of P increases while that of Q decreases till neutralisation.

(b) pH of P decreases while that of Q increases tillneutralisation.

(C) pH of both P and Q decrease.

(d) pH of both P and Q increase.

(iii) Which of the following acids is present in bee sting?
(a) Formic acid
(b) Acetic acid
(c) Citric acid
(d) Hydrochloric acid

(iv) Sting of ant can be cured by rubbing the affected area with soap because
(a) it contains oxalic acid which neutralises the effect of formic acid
(b) it contains aluminium hydroxide which neutralises the effect of formic acid
(c) it contains sodium hydroxide which neutralises the effect of formic acid
(d) none of these

(v) The pH of soil X is 7.5 while that of soil Y is 4.5. Which of the two soils, should be treated with powdered chalk to adjust its pH?
(a) X only
(b) Y only
(c) Both X and Y
(d) none of these

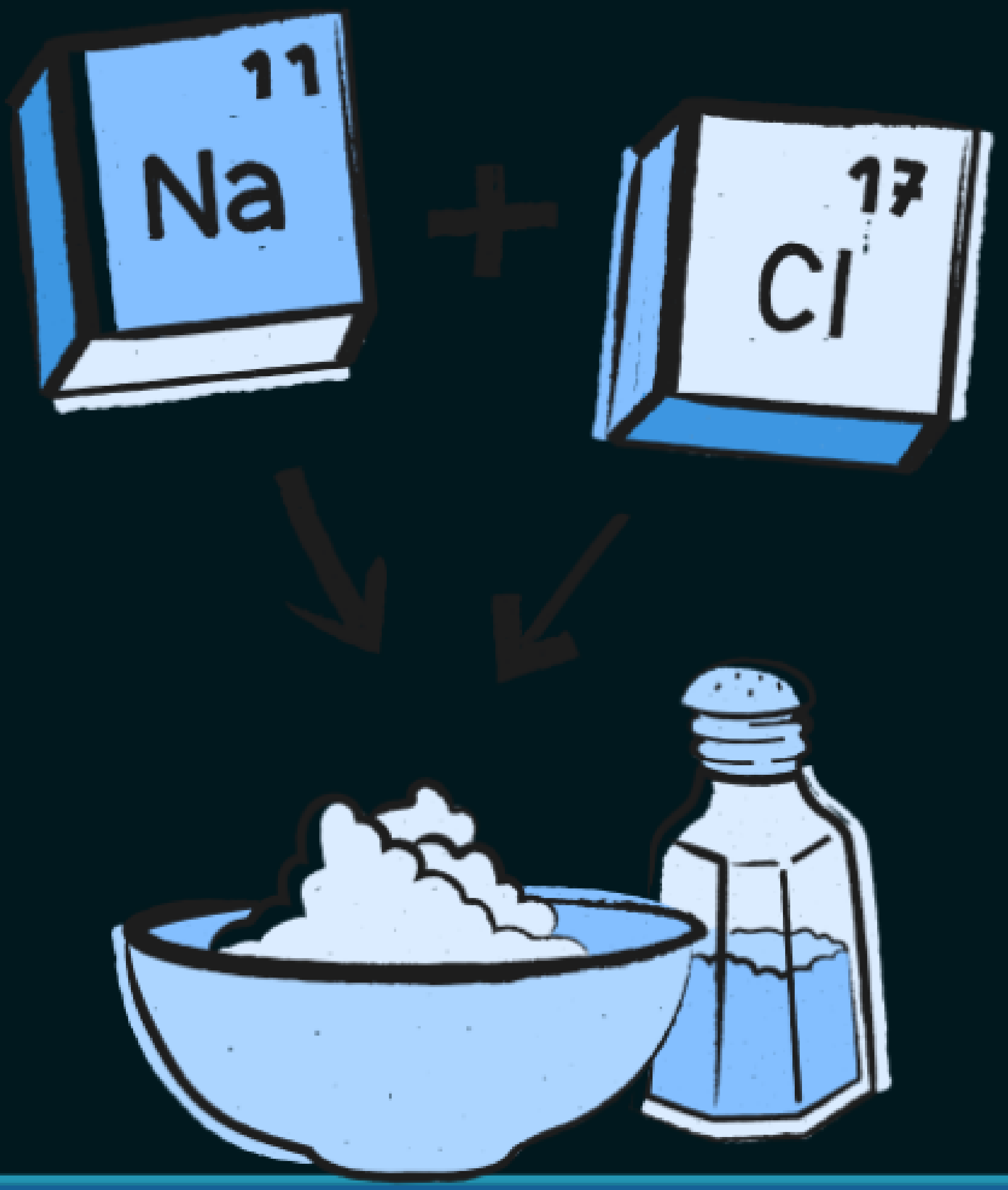
$CaCO_3 \rightarrow$ Basic

Salts

Salts

- A salt is an ionic compound formed by the neutralization reaction between an acid and a base.
- It consists of positively charged ions (cations) and negatively charged ions (anions), making the compound electrically neutral.

Acid + Base → Salt + Water



$HCl + NaOH \longrightarrow NaCl + H_2O$

Ionic Compound

$Na^{\oplus}$ $Cl^{\ominus}$

NaCl

Family of Salts

Salts having the same positive radical or negative radical belong to the same family.

1. **Sulphate salt family:** Salts containing SO_4^{2-} ion:
K_2SO_4, Na_2SO_4, $CaSO_4$, $MgSO_4$, $CuSO_4$

2. **Sodium salt family:** Salts containing Na^+ ion:
Na_2SO_4, $NaCl$, $NaNO_3$, Na_2CO_3

3. **Chloride salt family:** Salts containing Cl^- ion:
$NaCl$, NH_4Cl

$\underline{SA} + \underline{SB} \longrightarrow$ Neutral

$SA + WB \longrightarrow$ Acidic

$WA + SB \longrightarrow$ Basic

$HCl + NaOH$ $KOH + HNO_3$

Salt	Acid used	Base used	Nature	pH
$NaCl$	Strong acid	Strong base	**Neutral**	7
KNO_3	Strong acid	Strong base	**Neutral**	7
NH_4Cl → Acidic	Strong acid HCl	Weak base NH_4OH	**Acidic**	Less than 7
$CuSO_4$	Strong acid H_2SO_4	Weak base CuO	**Acidic**	Less than 7
CH_3COONa	Weak acid CH_3COOH	Strong base $NaOH$	**Basic**	More than 7
Na_2CO_3	Weak acid H_2CO_3	Strong base $NaOH$	**Basic**	More than 7
$NaHCO_3$	Weak acid H_2CO_3	Strong base $NaOH$	**Basic**	More than 7

SA $\rightarrow$ HCl, HNO_3, H_2SO_4

WA $\rightarrow$ CH_3COOH, Organic, H_2CO_3 (carbonic)

SB $\rightarrow$ $NaOH$, KOH

WB $\rightarrow$ $Mg(OH)_2$, $Ca(OH)_2$

Soch ke bataoo

Which of the given option represents a family of salts? [CBSE 2025]

(a) NaCl, Na_2SO_4, $CaSO_4$

(b) K_2SO_4, Na_2SO_4, $CaSO_4$

(c) $NaNO_3$, $CaCO_3$, Na_2CO_3

(d) $MgSO_4$, $CuSO_4$, $MgCl_2$

Soch ke bataoo

2020 (CBSE Board), 2021 (Sample Paper), 2023 (Practice Paper), 2024 (Sample Paper)

Q. An aqueous solution of a salt shows an orange colour when a drop of universal indicator is added to it. This salt is formed by:

(a) A strong acid and a strong base

(b) A weak acid and a weak base

(c) A strong acid and a weak base

(d) A weak acid and a strong base

$\boxed{\text{Orange}}$

Red → O

Acidic

Soch ke bataoo

2017, 2018 (Sample Paper), 2020, 2021 (Sample Paper), 2023 (CBSE Practice Paper), 2024 (Sample Paper)

Q. Complete the table by writing the nature of salt:

Acid	Base	Salt	Nature of Salt
HCl	$NaOH$	$NaCl$	N
H_2SO_4	NH_4OH	$(NH_4)_2SO_4$	A
CH_3COOH	$NaOH$	CH_3COONa	B
CH_3COOH	NH_4OH	CH_3COONH_4	N

Soch ke bataoo

Q. Study the following table and choose the correct option.

	Salt	Parent Acid	Parent Base	Nature of Salt
A	Sodium chloride	HCl	NaOH	Basic
B	Sodium carbonate	H_2CO_3	NaOH	Neutral
C	Sodium sulphate	H_2SO_4	NaOH	Acidic
D	Sodium acetate	CH_3COOH	NaOH	Basic

Soch ke bataoo

Q. Assertion (A): Sodium hydroxide reacts with hydrochloric acid to form sodium chloride and water.
Reason (R): The reaction between an acid and a base is called a neutralisation reaction.
[Sample Papers, 2011, 2015]

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, but R is not the correct explanation of A.
C. A is true, but R is false.
D. A is false, but R is true.

Common Salt

Common Name: Common Salt
Chemical Name: Sodium chloride (NaCl)
Chemical Formula: NaCl

- Common salt is made of sodium ions (Na^+) and chloride ions (Cl^-).
- It is obtained from sea water or salt mines.
- Sodium chloride is the most widely used salt in our daily lives.
- It's mainly used in food as a flavour enhancer and also acts as a preservative.

8 → Marks

50% (4)

50% (4)

Reaction for Formation:

It is formed by a neutralization reaction between an acid and a base:

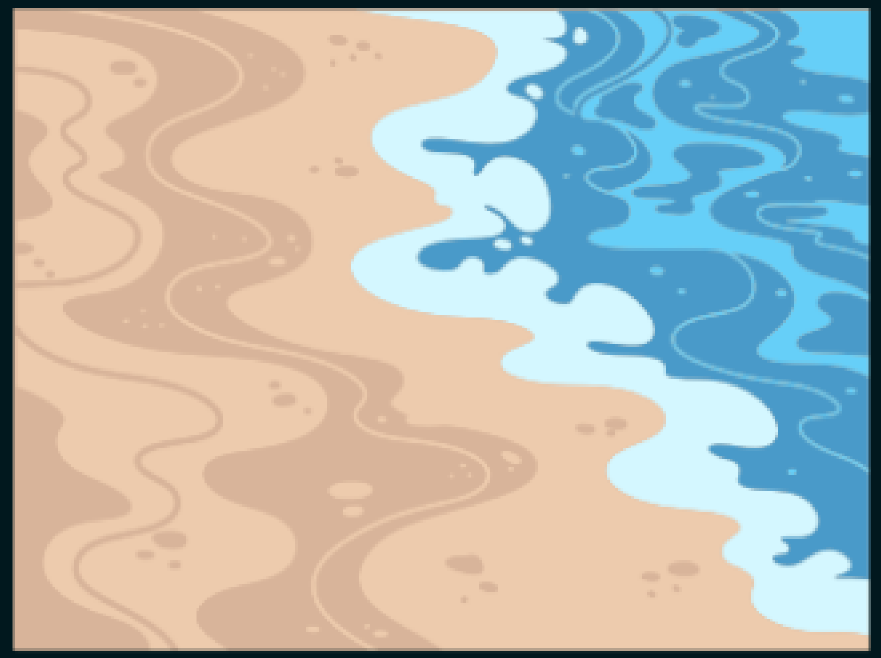
$$HCl + NaOH \rightarrow NaCl + H_2O$$

Uses of Common Salt:

- Used in cooking and seasoning food.
- Used for food preservation like pickling.
- Used for de-icing roads during winter.
- Raw material for various materials of daily use, such as sodium hydroxide, baking soda, washing soda, and bleaching powder

Sources of Common Salt

1. Seawater:



Sea water contains dissolved salts, mainly sodium chloride. Salt is obtained by evaporating sea water, leaving behind salt crystals.

2. Rock Salt

Rock salt is found in old seabeds formed by evaporation of ancient oceans. It is usually brown due to impurities and is mined like coal.

3. Inland Salt Lakes

Common salt is also obtained from inland salt lakes by evaporation.
Examples: Sambhar Lake, Rajasthan and Great Salt Lake, USA.

Soch ke bataoo

Q. What is the chemical name of rock salt? [2013, 2021]

(a) Sodium carbonate

(b) Sodium bicarbonate

(c) Sodium chloride

(d) Calcium chloride

Sodium Hydroxide

Chemical Formula: NaOH

Common name: Caustic soda

- It is a strong base and highly corrosive in nature.
- Sodium hydroxide dissolves in water to produce OH^- ions

Uses:

- **Used in manufacture of soaps and detergents**
- **Used in paper and artificial fibre industries**
- **Used for cleaning/degreasing metals**

Manufacturing of Sodium Hydroxide

- When electricity is passed through an aqueous solution of sodium chloride (brine), it decomposes to form sodium hydroxide.
- This process is called the chlor-alkali process.
- The name comes from the products formed:

Chlor → Chlorine gas
Alkali → Sodium hydroxide

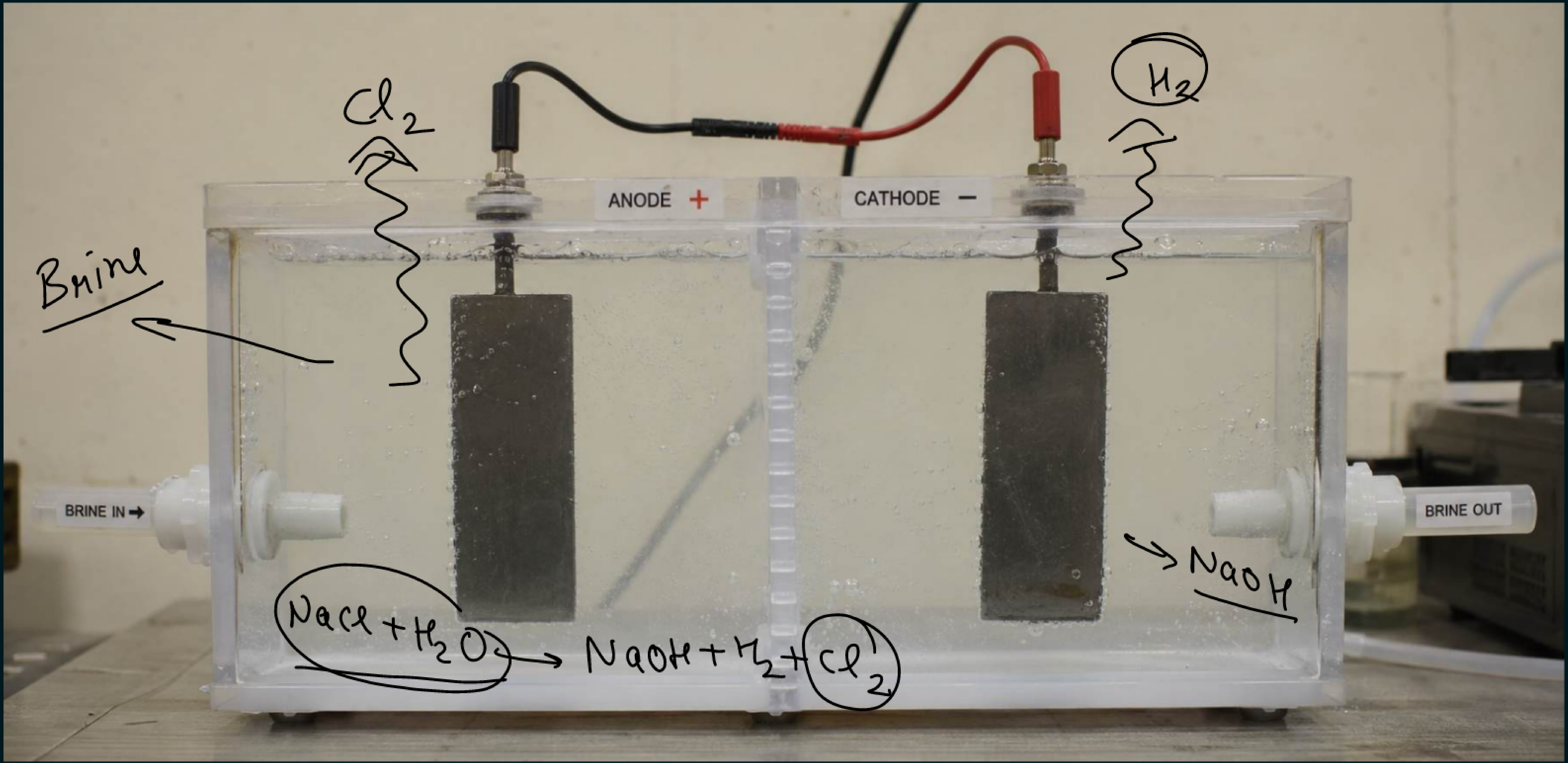
By-product

Main product

Chlor-alkali process reaction:

$$2NaCl(aq) + 2H_2O(l) \longrightarrow 2NaOH(aq) + Cl_2(g) + H_2(g)$$

Products of Chlor-Alkali Process and Their Uses



1. **Chlorine Gas (Cl_2) – At Anode**

Uses:

- Water treatment and swimming pools
- Manufacture of PVC
- Used as disinfectants
- Used in pesticides and chemicals

2. **Hydrogen Gas (H_2) – At Cathode**

Uses:

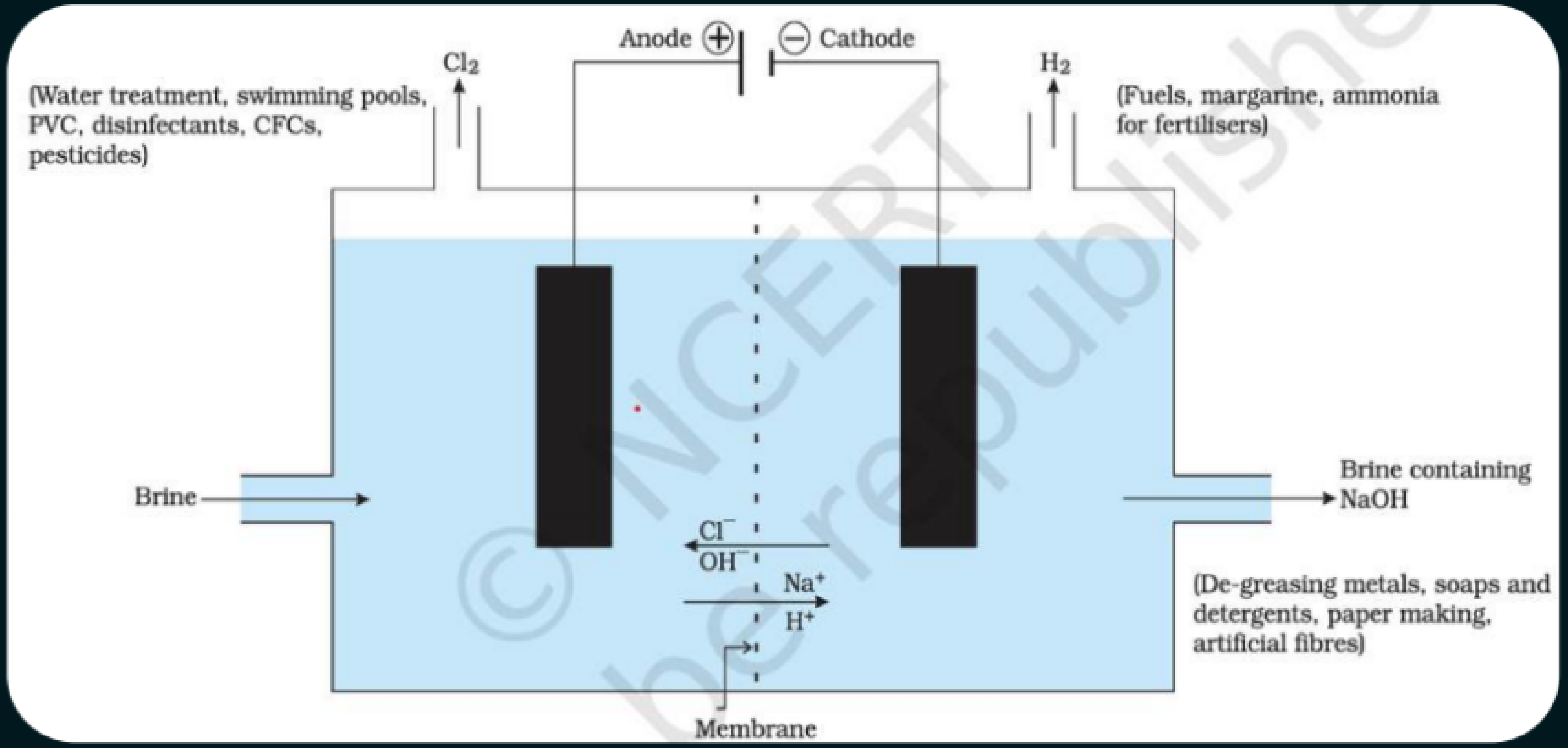
- Used as a fuel
- Used in hydrogenation of oils (margarine)
- Used in the manufacture of ammonia for fertilizers

NH_3

3. **Sodium Hydroxide (NaOH)**

Uses:

- Manufacture of soaps and detergents
- Paper and artificial fibre industries
- Used for cleaning and degreasing metals



Soch ke bataoo

Q. The chlorine produced during the electrolysis of brine solution is used in the manufacture of: [CBSE 2025]

(a) Ammonia

(b) Disinfectants

(c) Plaster of Paris

(d) Soap and detergents

Soch ke bataoo

Q. What are the products of chlor-alkali process? [2014, 2016, 2023]

(a) $NaOH$, HCl, CO_2

(b) $NaOH$, Cl_2, H_2

(c) $NaCl$, $NaOH$, Cl_2

(d) $NaHCO_3$, Cl_2, H_2

Soch ke bataoo

2017, 2018 (Sample Paper), 2020, 2021 (Sample Paper), 2023 (Practice Paper), 2024 (Sample Paper)

Q. In the electrolysis of brine, which gases are collected at the cathode and anode respectively?

(a) Hydrogen at cathode, chlorine at anode

(b) Oxygen at cathode, chlorine at anode

(c) Hydrogen at cathode, oxygen at anode

(d) Chlorine at cathode, hydrogen at anode

Soch ke bataoo

Q. Assertion (A): Sodium hydroxide is prepared by electrolysis of brine solution.
Reason (R): Chlorine and hydrogen are evolved in this process.
[2015, 2016, 2020, 2023]

Soch ke bataoo

2023 (CBSE Board – Set 2), 2024 (Sample Paper)

Q.The industrial process used for the manufacture of caustic soda involves electrolysis of an aqueous solution of compound 'X'. In this process, two gases 'Y' and 'Z' are liberated. 'Y' is liberated at cathode, and 'Z', which is liberated at anode, on treatment with dry slaked lime forms a compound 'B'.

Name X, Y, Z and B.

- X = Brine (aqueous solution of sodium chloride, NaCl)
- Y = Hydrogen gas (H_2)
- Z = Chlorine gas (Cl_2)
- B = Bleaching powder [$CaOCl_2$]

Soch ke bataoo

Q. Describe the method of preparation of Sodium Hydroxide from brine solution. Name the process and explain with a labelled diagram. Also write chemical equation and uses. [CBSE : 2017, 2019, 2020]

Soch ke bataoo

Q. To prepare a salad dressing, Parag adds a solution of sodium chloride in distilled water to vinegar. State what change will occur in the following:

(a) The pH of the vinegar

(b) The acidity of the vinegar

Soch ke bataoo

Q. Chlorine gas is prepared using electrolysis of brine solution. Write the chemical equation to represent the change. Identify the other products formed in the process and give one use of each.

Soch ke bataoo

When electricity is passed through an aqueous solution of sodium chloride (called brine), it decomposes to form sodium hydroxide. The process is called the chlor-alkali process because the products formed are chlorine and sodium hydroxide.

Q1: Write the chemical equation involved in this process.
Q2: What substances are formed at the anode and cathode in the chlor-alkali process?
Q3: What are the uses of chlorine?
Q4: Where is sodium hydroxide solution formed?
Q5: What are the uses of sodium hydroxide?

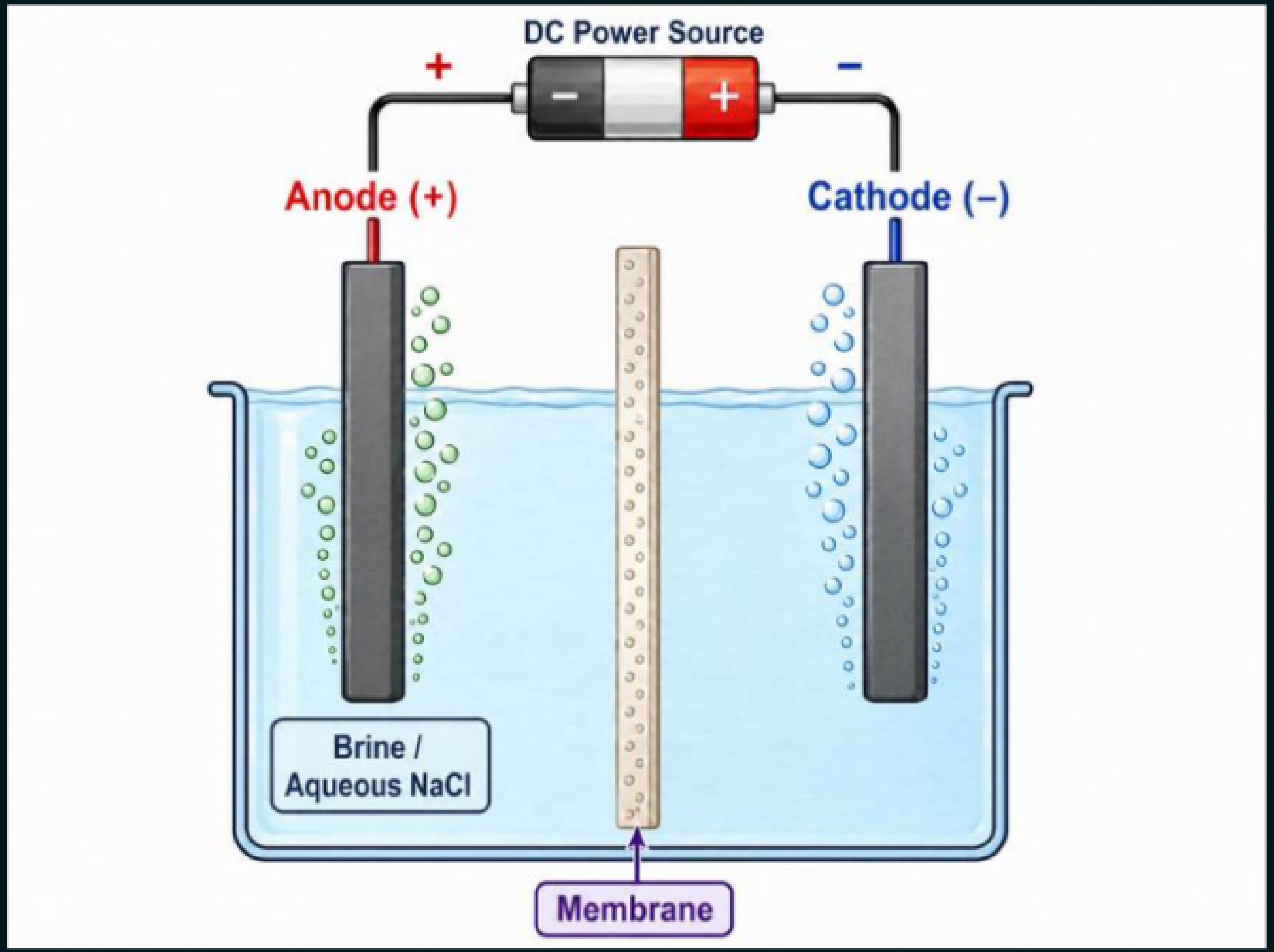
Soch ke bataoo

Q. The given diagram shows the chlor-alkali process.

(a) Name the gas evolved at the anode and cathode.

(b) Name the process that occurs. Why is it called so?

(c) Illustrate the reaction of the process with the help of a chemical equation.

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